BCA SEMESTER – V

INTRODUCTION TO MACHINE LEARNING

Theory & Machine Learning Lab (Using Python)

Extracted from the uploaded BCA syllabus

1. THEORY: INTRODUCTION TO MACHINE LEARNING

Course Code	Type of Course	Theory / Practical	Credits	Instruction / Week	Exam Duration	Formative Marks	Summative Marks	Total Marks
C5BCA1T1	Paper-1	Theory	04	04	3 hrs	20	80	100

Course Outcomes (COs)

- CO1. Understand the features of machine learning to apply on real world problems.
- CO2. Characterize the machine learning algorithms as supervised learning and unsupervised learning.
- CO3. Have a good understanding of the fundamental issues and challenges of machine learning.
- CO4. Learn the concepts in Bayesian analysis.
- CO5. Model evaluation and model selection.

Unit I – Introduction and Supervised Learning (15 hrs.)

Introduction, What Is Machine Learning? Examples of Machine Learning Applications, 12 Hours Supervised Learning: Learning a Class from Examples Vapnik-Chervonenkis (VC) Dimension, Probably Approximately Correct (PAC) Learning, Noise, Learning Multiple Classes, Regression, Model Selection and Generalization, Dimensions of a Supervised Machine Learning Algorithm.

Unit II – Bayesian Decision Theory (15 hrs.)

Bayesian Decision Theory: Introduction, Classification, Losses and Risks, Discriminant Functions, Utility Theory, Value of information, Bayesian Networks, Influence Diagrams, Association Rules.

Unit III – Parametric Methods (15 hrs.)

Parametric Methods: Introduction, Maximum Likelihood Estimation, Evaluating an Estimator: Bias and Variance, The Bayes' Estimator, Parametric Classification, Regression, Tuning Model Complexity: Bias Variance Dilemma, Model Selection Procedures.

Unit IV – Multivariate Methods and Clustering (15 hrs.)

Multivariate Methods: Multivariate Data, Parameter Estimation, Estimation of Missing Values, Multivariate Normal Distribution, Multivariate Classification, Tuning Complexity, Discrete Features, Multivariate Regression. Dimensionality Reduction: Introduction, Subset Selection, Principal Components Analysis, Factor Analysis, Multidimensional Scaling, Linear Discriminant Analysis.

Clustering: Introduction, Mixture Densities, k-Means Clustering, Supervised Learning after Clustering, Hierarchical Clustering, Choosing the Number of Clusters.

Text Book

1. Ethem Alpaydin, 2004, 'Introduction to machine Learning', PHI.

Reference Books

1. Tom M Mitchell, 1996, Machine Learning, McGraw Hill Publications.
2. Machine Learning: An Algorithmic Perspective by Stephen Marsland, Chapman and Hall/CRC.
3. Pattern Recognition and Machine Learning by Christopher M. Bishop, Springer.
4. Machine Learning by Tom Mitchell, McGraw Hill Education.

Formative Assessment for Theory

Assessment Occasion / Type	Marks
Internal Assessment Test 1	05
Internal Assessment Test 2	05
Assignment	10
Total	20

2. PRACTICAL: MACHINE LEARNING LAB (USING PYTHON)

Course Code	Type of Course	Theory / Practical	Credits	Instruction / Week	Exam Duration	Formative Marks	Summative Marks	Total Marks
C5BCA1P1	Paper-1	Practical	02	04	3 hrs	10	40	50

Course Outcomes (COs)

- CO1. Appreciate the importance of visualization in the data analytics solution.
- CO2. Apply structured thinking to understanding problems.
- CO3. Understand a very broad collection of machine learning algorithms and problems.
- CO4. Learn algorithmic topics of machine learning and mathematically deep enough to introduce the required theory.
- CO5. Develop an appreciation for what is involved in learning from data.

Programs List

5. Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples. Read the training data from a .CSV file.
6. For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm to output a description of the set of all hypotheses consistent with the training examples.
7. Write a program to demonstrate the working of the decision tree based on ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.
8. Write a program to implement the naïve Bayesian classifier for a sample training data set stored as a CSV file. Compute the accuracy of the classifier, considering few test data sets.
9. Assuming a set of documents that need to be classified, use the naïve Bayesian Classifier model to perform this task. Calculate the accuracy, precision, and recall for your data set.
10. Write a program to construct a Bayesian network considering medical data. Use this model to demonstrate the diagnosis of heart patients using standard Heart Disease Data Set.
11. Apply EM algorithm to cluster a set of data stored in a .CSV file.
12. Use the .CSV file data set for clustering using k-Means algorithm.
13. Write a program to implement k-Nearest Neighbour algorithm to classify the IRIS data set. Print both correct and wrong predictions.
14. Implement the non-parametric Locally Weighted Regression algorithm in order to fit data points. Select appropriate data set for your experiment and draw graphs.

Practical Assessment

Course Code	Type of Course	Theory / Practical	Credits	Instruction / Week	Exam Duration	Formative Marks	Summative Marks	Total Marks
C5BCA1P1	Paper-1	Practical	02	04	3 hrs	10	40	50

Source note: The theory and practical content above has been transcribed from the Machine Learning section of the uploaded BCA syllabus, corresponding to the syllabus pages containing the theory and Machine Learning Lab (Using Python).